3

(a)

Let $Z = X_{(k)}$. For each $i \in \{1, \dots, n\}$, define Z_{-i} to be the k-th largest observation among the $n-1$ values obtained by removing X_i. Since $k \leq n-1$, Z_{-i} is well defined. By the Efron-Stein inequality, we know that $Var(Z) \leq \sum_{i=1}^{n} E[Var^{(i)}(Z)]$, where $Var^{(i)}(Z) = Var(Z|X_{-i})$ and X_{-i} denotes all observations except X_i. Conditionally on X_{-i}, Z_{-i} is a constant because it does not depend on X_{-i}. We know that for any random variable Y and constant a:

$$\begin{aligned} E[(Y-a)^2] &= E[(Y - EY + EY - a)^2] \\ &= Var(Y) + (EY - a)^2 \\ &\geq Var(Y) \end{aligned}$$

Applying this conditionally with $Y = Z$ and $a = Z_{-i}$, we get $Var^{(i)}(Z) \leq E[(Z - Z_{-i})^2 | X_{-i}]$, and thus $Var(Z) \leq \sum_{i=1}^{n} E[(Z - Z_{-i})^2]$.

Let $\Delta = X_{(k)} - X_{(k+1)}$. Because observations are ordered in decreasing order, we know that $\Delta \geq 0$. We need to consider two cases:
Case 1: X_i is among the k largest observations
Then removing X_i makes the new largest kth value become the original $(k+1)$-th largest value, so $Z_{-i} = k+1$, and therefore $Z - Z_{-i} = X_{(k)} - X_{(k+1)} = \Delta$.
Case 2: X_i is not among the k largest observations
Then removing X_i does nothing to the k largest observations and $Z - Z_{-i} = 0$.

Now we sum over all observations with respect to the two cases:

$$\begin{aligned} \sum_{i=1}^{n} (Z - Z_{-i})^2 &= \underbrace{\Delta^2 + \cdots + \Delta^2}_{k \text{ terms}} + \underbrace{0 + \cdots + 0}_{n-k \text{ terms}} \\ &= k\Delta^2 \\ &= k(X_{(k)} - X_{(k+1)})^2 \\ \sum_{i=1}^{n} E[(Z - Z_{-i})^2] &= E[\sum_{i=1}^{n} (Z - Z_{-i})^2] \\ &= kE[(X_{(k)} - X_{(k+1)})^2] \end{aligned}$$

Then, by Efron-Stein, we finally get:

$$\begin{aligned} Var(X_{(k)}) &= Var(Z) \\ &\leq \sum_{i=1}^{n} E[(Z - Z_{-i})^2] \\ &= kE[(X_{(k)} - X_{(k+1)})^2] \end{aligned}$$

Thus, $Var(X_{(k)}) \leq kE[(X_{(k)} - X_{(k+1)})^2]$ for any $1 \leq k \leq n-1$.

(b)

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

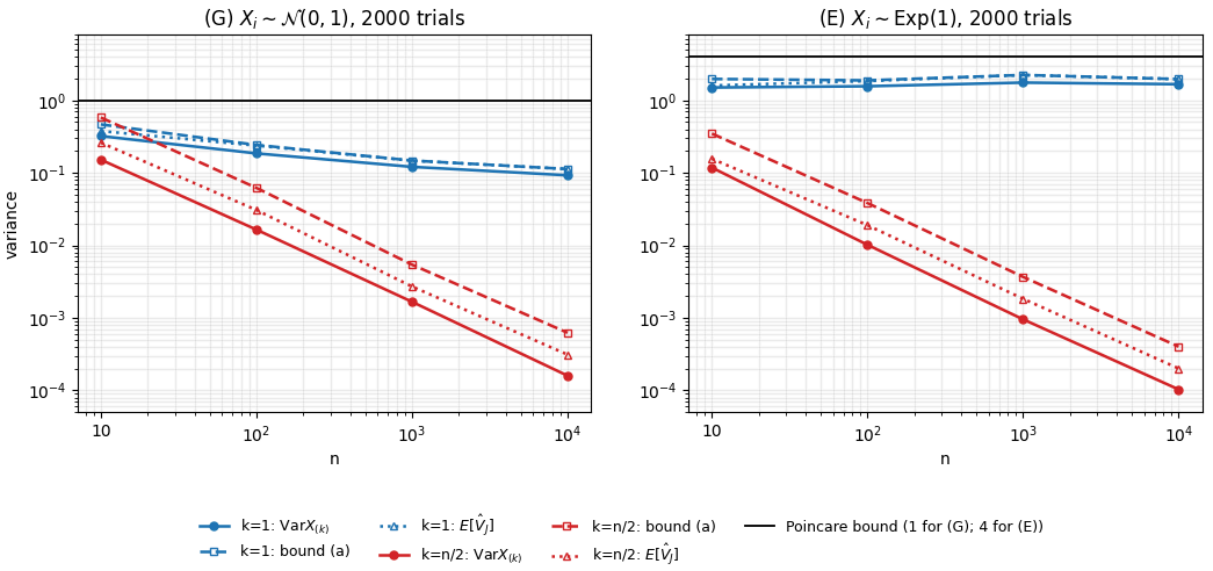
TRIALS = 2000
SIZES = [10, 100, 1000, 10000]
BATCH = 200
rng = np.random.default_rng(0)
rows = []

for dist in ('Gaussian', 'Exponential'):
    for n in SIZES:
        measurements = {kname: {'value': [], 'gap2': [], 'jackknife': []}
                        for kname in ('maximum', 'median')}
        for start in range(0, TRIALS, BATCH):
            count = min(BATCH, TRIALS - start)
            if dist == 'Gaussian':
                x = rng.standard_normal((count, n))
            else:
                x = rng.exponential(scale=1.0, size=(count, n))
            for kname, k in (('maximum', 1), ('median', n // 2)):
                p = np.partition(x, (n-k-1, n-k), axis=1)
                xk = p[:, n-k]
                xk1 = p[:, n-k-1]
                gap2 = (xk - xk1)**2
                jackknife = ((n-1)/n) * k * (1-k/n) * gap2
                measurements[kname]['value'].extend(xk.tolist())
                measurements[kname]['gap2'].extend(gap2.tolist())
                measurements[kname]['jackknife'].extend(jackknife.tolist())
        for kname, k in (('maximum', 1), ('median', n // 2)):
            values = np.asarray(measurements[kname]['value'])
            gap2 = np.asarray(measurements[kname]['gap2'])
            jack = np.asarray(measurements[kname]['jackknife'])
            v = values.var(ddof=1)
            b = k * gap2.mean()
            j = jack.mean()
            rows.append({'distribution': dist, 'n': n, 'k': k,
                         'statistic': kname, 'variance': v, 'bound_a': b,
                         'mean_jackknife': j,
                         'relative_sd_jackknife': jack.std(ddof=1)/j,
                         'bound_to_variance': b/v,
                         'jackknife_to_variance': j/v})
```

```
df = pd.DataFrame(rows)
```

In [11]:
```
fig, axes = plt.subplots(1, 2, figsize=(12.5, 5.0), sharex=True)
colors = {'maximum': 'tab:blue', 'median': 'tab:red'}
style = [('variance', '-', 'o', r'Var$X_{(k)}$'), ('bound_a', '--', 's', 'bound (a)
         ('mean_jackknife', ':', '^', r'$E[\hat{V}_J]$')]
for ax, dist in zip(axes, ('Gaussian', 'Exponential')):
    sub = df[df.distribution == dist]
    for kname in ('maximum', 'median'):
        data = sub[sub.statistic == kname]
        for col, ls, marker, label in style:
            ax.plot(data.n, data[col], ls, marker=marker, linewidth=1.8,
                    markersize=5, color=colors[kname],
                    label=f"k={'1' if kname == 'maximum' else 'n/2'}: {label}",
                    markerfacecolor='none' if col != 'variance' else colors[kname])
    pc = 1 if dist == 'Gaussian' else 4
    ax.axhline(pc, color='black', linewidth=1.3, linestyle='-',
               label='Poincare bound (1 for (G); 4 for (E))')
    ax.set_xscale('log')
    ax.set_yscale('log')
    ax.set_xticks(SIZES, ['10', r'10$^2$', r'10$^3$', r'10$^4$'])
    if dist == 'Gaussian':
        ax.set_title(r'(G) $X_i \sim \mathcal{N}(0, 1)$' + f', {TRIALS} trials')
    else:
        ax.set_title(r'(E) $X_i \sim \mathrm{Exp}(1)$' + f', {TRIALS} trials')
    ax.set_xlabel('n')
    ax.grid(True, which='both', alpha=0.22)
    ax.set_ylim(5e-5, 8)
axes[0].set_ylabel('variance')
handles, labels = axes[0].get_legend_handles_labels()
fig.legend(handles, labels, loc='lower center', bbox_to_anchor=(0.5, -0.06),
           ncol=4, fontsize=9, frameon=False)
fig.subplots_adjust(bottom=0.23, wspace=0.19)
plt.show()

print(df.to_string(index=False, float_format=lambda z: f'{z:.6g}'))
```



```
distribution     n    k statistic    variance     bound_a  mean_jackknife  relative_
sd_jackknife  bound_to_variance  jackknife_to_variance
    Gaussian    10    1   maximum    0.320902    0.464946        0.376606
1.65096            1.44887               1.17359
    Gaussian    10    5    median    0.151616     0.57883        0.260473
1.8184           3.81774              1.71798
    Gaussian   100    1   maximum    0.185226    0.242309        0.237487
1.97451            1.30818               1.28215
    Gaussian   100   50    median     0.01647   0.0619844       0.0306823
2.07567            3.76348               1.86292
    Gaussian  1000    1   maximum    0.120886    0.147417        0.147122
1.92604            1.21947               1.21703
    Gaussian  1000  500    median  0.00166544  0.00541575      0.00270517
2.39296            3.25185                1.6243
    Gaussian 10000    1   maximum   0.0924339    0.112739        0.112717
1.99764            1.21967               1.21943
    Gaussian 10000 5000    median 0.000159376 0.000623341     0.000311639
2.10235            3.91114               1.95538
 Exponential    10    1   maximum     1.50107     1.96738         1.59358
2.30665            1.31065               1.06163
 Exponential    10    5    median    0.118025    0.347289         0.15628
2.05495             2.9425               1.32412
 Exponential   100    1   maximum     1.56115     1.88037         1.84295
2.15488            1.20448               1.18051
 Exponential   100   50    median   0.0101714   0.0383104       0.0189637
2.08279             3.7665               1.86442
 Exponential  1000    1   maximum      1.7531     2.23194         2.22748
2.60704            1.27314               1.27059
 Exponential  1000  500    median 0.000954521  0.00367163      0.00183398
2.05535            3.84657               1.92136
 Exponential 10000    1   maximum     1.66509     1.95843         1.95804
2.16832            1.17617               1.17594
 Exponential 10000 5000    median 0.000103092 0.000401382     0.000200671
2.28592            3.89344               1.94652
```

i.

Gaussian max: Variance decreases slowly as n increases at a rate of $\frac{1}{\log n}$. Both bound (a) and the expected jackknife follow the same order of decrease. Poincare bound remains constant at 1.

Gaussian median: Variance decreases slowly as n increases at a rate of around $\frac{1}{n}$. Both bound (a) an the expected jackknife follow the same order of decrease but overestimate the variance.

Exponential max: Variance approaches a positive constant rather than decreasing to zero. Bound (a) and the expected jackknife approach 2, so they have the correct constant-order dependence.

Exponential median: Variance decreases slowly as n increases at a rate of around $\frac{1}{n}$. Both bound (a) an the expected jackknife follow the same order of decrease.

The Poincare bound doesn't capture these rates because for the k-th largest order statistic, $f(x) = x_{(k)}$. Whenever all coordinates are distinct, exactly one coordinate determines the k-th largest value. Hence, the gradient only has one entry equal to 1 and all other entries are equal to 0. Therefore, $\|\nabla f(x)\|_2^2 = 1$ almost everywhere, giving $E[\|\nabla f(X)\|_2^2] = 1$, and

any upper bound of the form $CE[\nabla f(X)\|_2^2] = C$ is constant in n. Consequently, it can't capture the decreasing value for either median or max Gaussian.

ii. At $n = 10,000$: | Distribution | k | Bound (a) factor | Jackknife factor |-|-|-|-| | Gaussian | 1 | 1.22x | 1.22x | | Gaussian | 5,000 | 3.91x | 1.96x | | Exponential | 1 | 1.18x | 1.18x | | Exponential | 5,000 | 3.89x | 1.95x |

We can see that bound (a) overestimates by around 20% for $k = 1$ and 4 times for $k = 5,000$ while jackknife overestimates by around also 20% for $k = 1$ but only 2 times for $k = 5,000$.

iii. No, the simulation indicates that the ratio goes to around 2. For exponential order statistics, the adjacent spacing satisfies $\Delta = X_{(k)} - X_{(k+1)} \sim \text{Exp}(k)$. We already know that the jackknife is a constant multiple of Δ^2, so its relative standard deviation also equals that of Δ^2. For an exponential random variable with rate k, $E[\Delta^2] = \frac{2}{k^2}$, $E[\Delta^4] = \frac{24}{k^4}$. Consequently, $Var(\Delta^2) = E[\Delta^4] - (E[\Delta^2])^2 = \frac{24}{k^4} - \frac{4}{k^4} = \frac{20}{k^4}$. Therefore, the ratio is now equal to $\frac{\sqrt{20/k^4}}{2/k^2} = \sqrt{5} \sim 2.2$. Thus, for exponential observations, the jackknife's relative standard deviation is exactly $\sqrt{5}$ for every sample size and every eligible k. The Gaussian results are also consistent with this, because the appropriately rescaled adjacent order-statistic gaps approach exponential distributions.

No, we should not use the jackknife to estimate the variance of a sample median since it is unreliable for estimating variance. Its expected value is approximately twice the actual median variance in the simulation, so it has substantial upward bias. Second, its relative standard deviation does not approach zero. Even with a very large sample, one observed adjacent spacing can produce a highly variable variance estimate.